



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam: : Trimester: Summer - 2024

Course Code: CSE 4889/CSE 489, **Course Title:** Machine Learning, **Sec:** A

Total Marks: 30

Duration: 1:45 hours

Please answer all the questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Question 1:

5+5=10

- A. Discuss different methods that commonly used in data preprocessing.
- B. Explain the Multivariate Linear Regression with following example.

X_1	X_2	y_1	y_2
9	62	10	100
8	58	12	110
7	64	12	105

Question 2:

5+5=10

- A. Explain K-Nearest-Neighbors classifier with its advantages and disadvantages.
- B. Find the decision tree using CART algorithm from the Table 1.

Table 1: Playing Dataset

Eye-color	Married	Sex	Hair-length	Class
Brown	Yes	Male	Long	Football
Blue	Yes	Male	Short	Football
Brown	Yes	Male	Long	Football
Brown	No	Female	Long	Netball
Brown	No	Female	Long	Netball
Blue	No	Male	Long	Football
Brown	No	Female	Long	Netball
Brown	No	Male	Short	Football
Brown	Yes	Female	Short	Netball
Brown	No	Female	Long	Netball
Blue	No	Male	Long	Football
Blue	No	Male	Short	Football

Question 3:

5+5=10

- A. What is attribute bagging? Write the difference between RandomForest and Bagging techniques.
- B. Explain Adaboost classifier with its algorithm.